

Sample preparation protocol for targeted mass spectrometry analysis of pRabs, total Rabs, LRRK2-pS910, pS935 and total LRRK2

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Abstract

Leucine rich repeat kinase2 (LRRK2) is one of the most commonly implicated kinase and mutations that hyperactivate kinase activity is seen in familial Parkinson's disease. LRRK2 phosphorylates a sub-set of small Rab GTPase proteins within their Switch-II motif. We recently described a multiplexed targeted mass spectrometry assay to identify and quantify both phosphorylated Rab proteins and well known LRRK2 phosphorylation sites such as LRRK2-pS910/pS935(PMID: 33367571). Here we describe a detailed protocol for the sample preparation of cells/tissues that are derived from LRRK2 mutation knock-in models. We outline a simple Triton-X 100 or SDS lysis that are straightforward to capture solubilized proteins amenable for both immunoblotting and mass spectrometry analysis. A stepwise protocol to buffer exchange Triton-X 100/SDS using S-Trap columns and on-column trypsin digestion. We also describe an optimized condition to conjugate required antibodies for the selective enrichment of pRab's and LRRK2 phosphopeptides and total LRRK2 peptide including the protocol for peptide immunoprecipitation and subsequent Stage tip-based desalting of the samples prior to targeted mass spectrometry analysis. We provide a cautionary step's in each sections of the protocol for a reproducible sample preparation. We believe this protocol is an easy handout to study LRRK2 activity both from cells/tissues including samples from human neutrophils.

1) Materials

1.1) Consumables:

- 1.1.1) 1.5 ml protein low bind Eppendorf tubes (Eppendorf™ #022431081)
- 1.1.2) Marker pen
- 1.1.3) Pipette set (1 ml, 200 µl, 100 µl, 20 µl, 10 µl)
- 1.1.4) Pipette tips low binding (1 ml, 250 µl, 10 µl, Star labs Bevelled tips refill # S1111-3700, S1111-1706, S1111-6700)
- 1.1.5) PPE kit (Lab coat, gloves, safety glasses)
- 1.1.6) Dry ice
- 1.1.7) Liquid Nitrogen
- 1.1.8) Ice bucket
- 1.1.9) 1.5 ml eppendorf tubes rack
- 1.1.10) 96 well plate- clear (Geneier Bio-one #655101)
- 1.1.11) 2 ml tubes (Axygen™ MCT2000)

- 1.1.12) 16-gauge needle (# Z261378. Sigma Aldrich)
- 1.1.13) Spray duster (Qconnect #KFO4499)
- 1.1.14) X100 20 mL Amber Glass EPA Vial (Thermo Scientific™ EPA Screw Vial Assembled Kit, 20 mL amber glass EPA vial with cap and seal. Fisher Scientific # 11543750)
- 1.1.15) X72 40 mL Amber class EPA vial W Cap and seal (Thermo Scientific™ EPA, TOC, and Scintillation Vials & Closures. Fisher Scientific # 12418656)
- (Note: Prepare all stock and working reagents in these amber vials to store as per the protocol)
- 1.1.16) Millipore pH Strips (VWR # 1.09584.0001)
- 1.1.17) BCA protein assay kit (Pierce # 23225)
- 1.1.18) PTFE- O rings (To place the stage-tip into the Eppendorf tubes. Generally, you could get from NEST group desalting columns and re use them <https://www.nestgrp.com/>)
- 1.1.19) S-Trap mini columns (<https://www.protifi.com/>)
- 1.1.20) Sep-Pak Vac 1cc (50 mg) tC18-Cartridges (Waters # WAT054960)
- 1.1.21) Evotips (EvoSep #EV2001)
- 1.1.22) EvoSep 8cm analytical column (EvoSep #EV-1064)
- 1.1.23) EvoSep 15cm performance analytical column (EvoSep #EV-1118)
- 1.1.24) Easy-Spray PepMap RSLC C18 2um, 50cmx75um (Thermo Fisher Scientific #ES802)
- 1.1.25) Heavy stable synthetic peptides. Custom design by JPT peptide technologies (<https://www.jpt.com/>)
- 1.1.26) Pierce retention time calibration mix (Thermo Fisher Scientific #88321)

1.2) Reagents:

- 1.2.1) SDS Lysis Buffer: Final 2% (by mass) SDS in 100 mM Triethylammonium bicarbonate pH 8.5 (TEABC, this is the natural pH of this buffer and made from a 1 M TEABC stock purchased from Sigma Catalogue number# T7408-500 mL), 1 mM sodium orthovanadate, 50 mM NaF, 10 mM β - glycerophosphate, 5 mM sodium pyrophosphate, 1 μ g/ml microcystin-LR, and complete EDTA-free protease inhibitor cocktail (Roche)
- 1.2.2) 1% Triton-x 100 Lysis buffer: Final 1% (by mass) SDS in 100 mM Triethylammonium bicarbonate pH 8.5 (TEABC, this is the natural pH of this buffer and made from a 1 M TEABC stock purchased from Sigma Catalogue number# T7408-500 mL), 1 mM sodium orthovanadate, 50 mM NaF, 10 mM β - glycerophosphate, 5 mM sodium pyrophosphate, 1 μ g/ml microcystin-LR, and complete EDTA-free protease inhibitor cocktail (Roche)

- 1.2.3) Tris (2-carboxyethyl) phosphine (TCEP) (Sigma Aldrich # 75259-10G). (Note: Prepare and store 10 µl aliquots of 1 M TCEP in Milli-Q H₂O. Prior to use dilute the 1 m TCEP solution 10 x in 300 mM TEABC to generate a stock solution of 0.1 M TCEP in 300 mM TEABC).
- 1.2.4) Ortho-Phosphoric acid 85% (by vol) (Sigma #5438280100). (Note: Prepare 12% (by vol) stock aqueous phosphoric acid by diluting in water and store in 4°C)
- 1.2.5) S-Trap protein binding buffer (90% (by vol) LC grade methanol containing a final concentration of 100 mM TEAB, pH 7.1, made from a 1 M TEABC stock purchased from Sigma Aldrich # T7408-500 mL)
- 1.2.6) Trypsin from bovine pancreas TPCK treated (Sigma Aldrich #V5111). (Note: Prepare a stock at 5 µg/µl of 50 µl volume vials in 0.05% Acetic acid and store in -20 freezer)
- 1.2.7) CBQCA Protein Quantitation Kit (Thermo Fisher Scientific #C-667)
- 1.2.8) 10X Immunoaffinity purification (IAP) buffer. Prepare 10X IAP buffer containing 500 mM MOPS, 100 mM Na₂HPO₄ and 500 mM NaCl as a stock and dilute it with Milli-Q H₂O to achieve 1X as working solution and store at 4°C (MOPS 3-(N-Morpholino) propanesulfonic acid, 4-Morpholinepropanesulfonic acid (Sigma Aldrich # 69947-500G; Sodium phosphate dibasic Na₂HPO₄ dibasic (Sigma Aldrich # S7907-500G); Sodium chloride (Sigma Aldrich # S7653-1KG)
- 1.2.9) Gibco™ PBS, pH 7.4 (Fisher Scientific # 11593377)
- 1.2.10) Protein A/G beads (<https://mrcppureagents.dundee.ac.uk/>)
- 1.2.11) Pan-pRab8 antibody (Abcam #MJF-R20, ab230260)
- 1.2.12) LRRK2 pS910 antibody (UDD1 <https://mrcppureagents.dundee.ac.uk/>)
- 1.2.13) LRRK2 pS935 antibody (UDD2 <https://mrcppureagents.dundee.ac.uk/>)
- 1.2.14) LRRK2 8G10 antibody (8G10 <https://mrcppureagents.dundee.ac.uk/>)
- 1.2.15) Methanol (VWR # 1.06035.2500)
- 1.2.16) Acetic acid 100% (Sigma #5438080100)
- 1.2.17) LC-MS grade Acetonitrile (VWR # 1.00030.2500)
- 1.2.18) Iodoacetamide (Sigma # I1149)
- 1.2.19) LC grade Formic acid (Sigma # 695076)
- 1.2.20) Trifluoroacetic acid (Sigma # 302031) (Note: Prepare and store 20% (by vol) aqueous TFA stock at 4°C)
- 1.2.21) Empore C18 disks, 47 mm (CDS analytical #2215) (Note: Prepare a single layer with 16-gauge needle and pass it with spray duster into the 250 µl tip for 0.1 to 5 µg of peptide)

amount. For more than 5 ug use 2 or 3 layers of C18 material. Refer Figure 1 -see below for Stage-tip assembly).

- 1.2.22) 20% (by mass) aqueous SDS stock
- 1.2.23) LC buffer (0.1% (by vol) Formic acid in 3% (by vol) Acetonitrile)
- 1.2.24) Solvent-A1 (0.1% (by vol) TFA)
- 1.2.25) Solvent-A2 (0.1% (by vol) Formic acid)
- 1.2.26) Solvent-B1 (50% (by vol) acetonitrile 0.1% (by vol) TFA)
- 1.2.27) Solvent-B2 (60% (by vol) acetonitrile 0.1% (by vol) Formic acid)

1.3) Equipment:

- 1.3.1) Pulveriser kit (<https://cellcrusher.com/>)
- 1.3.2) -80°C deep freezer, -20°C freezer and 4°C fridge
- 1.3.3) Benchtop centrifuge (VWR)
- 1.3.4) Milli-Q water system
- 1.3.5) Orbital shaker
- 1.3.6) pH meter
- 1.3.7) Ice rack
- 1.3.8) Plate reader for Protein quantification (BioTek Epoch)
- 1.3.9) Diagenode Bioruptor plus sonication system
- 1.3.10) Eppendorf Thermomixer
- 1.3.11) Thermo Savant Speedvac system (#SPD140DDA)
- 1.3.12) 1.5 ml tube floaters
- 1.3.13) PHERAstar FSX
- 1.3.14) Branson water bath sonicator
- 1.3.15) Dionex RSLC 3000 nano-LC system
- 1.3.16) Dionex RSLC 3000 LC system for Offline fractionation with Auto sampler or Fraction collector, micro pump and VWD detector
- 1.3.17) Orbitrap Fusion Lumos Tribrid or QExactive HF-X or Orbitrap Exploris 480 or Orbitrap Exploris 240 Mass spectrometer(s)
- 1.3.18) Thermo Savant Speed vac system (#SPD140DDA)
- 1.3.19) Nanodrop 1000 (Thermo Fisher Scientific)
- 1.3.20) Rubber bulb # Fisher brand™ Rubber Pipette Bulb# 12446180.

2) Protocol:

2.1) Protocol for lysate preparation from mice tissues:

- 2.1.1) Snap freeze mouse tissue immediately after isolation in liquid nitrogen and store at -80 °C.
- 2.1.2) Pulverize frozen tissue in liquid nitrogen to a fine powder and snap freeze immediately in 2 ml tubes. This is done using cell crusher kit.
(Note: Maintain all Pulverizer units in liquid nitrogen including spatula. After use wash the pulveriser with a tap water and clean thoroughly several times with 70% (by vol) ethanol)
- 2.1.3) Weigh pulverized tissue powder on a weighing balance (Note1: 5-7% of the wet weight of tissue should correspond to a protein amount of ~ 5 to 7 mg protein/100 mg of wet weight) (Note2: Keep samples on dry ice during and after weighing. Take a 1.5 ml tube check the weight on weighing balance and subsequently tare the weight. Now take small scoop of sample using pre-chilled fine spatula to weigh the sample).
- 2.1.4) Immediately add 10 times of the weight of the tissue of 2% (by mass) SDS lysis buffer (Note: For e.g. 100 mg of tissue powder please add 1000 µl of lysis buffer).
- 2.1.5) Put samples on an orbital shaker in cold room at 1000 rpm for 15 min.
- 2.1.6) Boil samples at 95°C for 10 min and allow to cool.
- 2.1.7) Sonicate samples using a Diagenode Bioruptor (use it at high energy for 10 cycles (30sec-ON/30Sec-Off)).
- 2.1.8) Clarify lysate by centrifuging at 20,800g for 30 min at 4°C.
- 2.1.9) Measure protein amount using the Bicinchoninic acid assay (BCA) method in triplicate at 1 to 10 (cells) or 1 to 20 (tissues) dilutions-repeat analysis if readings are not close.
- 2.1.10) Take 300 µg of protein amount for Mass spectrometry analysis. (Note: In case of mice brain tissues the minimum recommended lysate, amount is 1mg to identify and quantify endogenous pRab proteins and LRRK2 pS910/pS935 and total LRRK2 peptide).

2.2) Protocol for lysate preparation from cells:

- 2.2.1) For cells such as A549 or HEK293 or Mouse embryonic fibroblasts (MEFs): Culture them in 6cm or 10cm dish to a ~70% confluency and complete the required treatments such as MLI2 treatment for about 90 min.
- 2.2.2) Harvest cells by washing with 1X plain DMEM media without any serum and aspirate the media completely. Lyse cells using 1% Triton-X 100 lysis buffer. Add 150 µl to a 6cm dish or 350 µl to a 10cm dish. Scrape cells using cell scraper and collect the lysate using 1 ml non-autoclaved tips into a protein lo binding tubes.
- 2.2.3) Keep cells on ice for 30 min and centrifuge at 18,000g for 20 min at 4°C.

- 2.2.4) Estimate protein amounts using BCA assay.
- 2.2.5) Take 300 µg of protein amount for Mass spectrometry analysis (cell lysate or tissue lysate).

2.3) Protocol for SDS-clean up and S-Trap assisted trypsin digestion:

- 2.3.1) Aliquot 300 µg of protein amount in 1.5 ml Protein lo-binding tubes. Perform reduction by adding 1 in 10 dilution of a solution of 0.1 M TCEP dissolved in 300 mM TEABC to bring final concentration of TCEP to 10 mM.
- 2.3.2) Incubate on a Thermomixer for 30 min at 60°C temperature with a gentle agitation at 1000 rpm.
- 2.3.3) Bring tubes to **room temperature** and add a one in ten dilution of freshly prepared 0.4 M iodoacetamide dissolved in water (**Note: It is critical that the samples are at room temperature prior to addition of iodoacetamide**).
- 2.3.4) Incubate in dark on a Thermomixer at room temperature for about 30 min with a gentle agitation at 1000 rpm.
- 2.3.5) Quench alkylation by addition of a 1 in 10 dilution of 0.1 M TCEP dissolved in 300 mM TEABC to bring final concentration of TCEP to 10 mM (**Note: At this stage sample can be stored in -80 freezer. If samples are not quenched after alkylation proceed immediately to the next steps below**).
- 2.3.6) Incubate on a Thermomixer for 20 min at room temperature with a gentle agitation at 1000 rpm.
- 2.3.7) Add SDS to a final concentration of 5% (by mass) from 20% (by mass) SDS stock.
(**Note: The lysate is already in 2% (by mass) SDS, so supplement with a stock of 20% (by mass) SDS to bring the final SDS concentration to 5% (by mass). For lysates with 1% Triton-x 100, it is important to add final 5% (by mass) SDS to the samples to ensure complete denaturation and subsequent capturing of fine protein particles by S-Trap columns**).
- 2.3.8) Add a 1 in 10 dilution of 12% (by vol) phosphoric acid into the sample to make a final concentration of ~1.2% (by vol) phosphoric acid (**Note: For example, 5 µl for 50 µl of lysate**).
- 2.3.9) Dilute the sample 7 times in S-Trap wash buffer (90% (by vol) methanol in 0.1 M TEAB pH 7.1 v/v) (For example, if sample volume is 50 µl then add 300 µl of S-Trap wash buffer ((90% (by vol) methanol in 0.1 M TEAB pH 7.1 v/v)) (**Note: If the lysate volume is more than 50 µl, scale S-Trap wash buffer accordingly i.e. 7 times the volume of the lysate**).
- 2.3.10) Prepare an S-Trap mini column in a 2 ml tube.
- 2.3.11) Add the diluted protein mixture from section 2.3.9 to the column.

- 2.3.12) Centrifuge briefly to capture the protein particles at 1000g for 1 min at room temperature.
(Note: All centrifugation steps should be at room temperature. If you observe any clogging then increase the centrifugation speed up to a maximum of 4,000g or increase the centrifugation time up to 2-4 minutes. If the total volume (Lysate +S-Trap buffer mixture is >500 µl then add it to the column multiple times until the sample passes through the column and discard flowthrough).
- 2.3.13) Wash column with 400 µl of S-Trap buffer a total of 4 times (spin 1000g for 1 min between washes) (Note: The protein remains bound on the column and SDS and buffer components that effect trypsin digestion are removed).
- 2.3.14) Move the S-Trap column to a clean 2 ml tube for digestion.
- 2.3.15) Add a 160 µl solution of freshly dissolved trypsin containing 15 µg of trypsin dissolved in 100 mM TEAB (1:20): (Note: Prepare and store Trypsin stocks in Milli-Q water at 5µg/µl of 50µl vials and store in -20 freezer).
- 2.3.16) Centrifuge briefly at 200g for 1 min.
- 2.3.17) Collect flow through and reapply the trypsin solution back onto the column being careful to avoid air bubbles.
- 2.3.18) Cap the tubes and incubate at 47°C without shaking for 1.5 hour. (Note do not shake as this causes bubbles and damage the column).
- 2.3.19) Incubate for another 2 h at room temperature.
- 2.3.20) Add 80 µl of 50 mM TEAB then spin to elute and place the eluate in a new 1.5 ml Eppendorf tube termed “eluate tube”.
- 2.3.21) Next, add 80 µl of 0.15% (by vol) Formic Acid and spin to elute. Also add this eluate to the “eluate tube”.
- 2.3.22) Finally, add 80 µl of 50% (by vol) Acetonitrile in 0.15% (by vol) formic acid and spin to elute. Also add this eluate to the “eluate tube” (Note 3 eluates should have been added to the eluate tube).
- 2.3.23) Take 1-2 µl of the combined eluate, vacuum dry and inject on MS to verify the digestion efficiency (Note: Analyse data with a 70 min gradient run on QE HF-X or Orbitrap Lumos mass spectrometer in a FT-FT-HCD mode. Search data with Proteome Discoverer 2.1 or 2.4 version. Determine the digestion efficiency by plotting number of missed cleavages. Zero missed cleavages should be >75% and single missed cleavages should be between 20-23%).
- 2.3.24) Estimate peptide amounts using CBQCA assay kit and take equal amounts of peptide digest for each sample and Vacuum dry using SpeedVac (Note: Aliquot 5 µg of peptide digest separately for total Rab analysis).

3.1) Pan-pRab, UDD1, UDD2 and 8G10 antibodies conjugation to Protein A/G beads:

- 3.1.1) Take required amount of Protein A/G beads and wash with a total of 1 ml containing, slurry volume + reminder is PBS wash volume (1X PBS buffer). Centrifuge at 2,000g for 3 min (e.g. For 24 samples with 2 rounds of immunoprecipitations 480 µl of bead slurry is required. 10µl of slurry/sample for one round of immunoprecipitation).
(Note: Let the beads settle down for 2 min before aspirating the supernatant.)
- 3.1.2) Repeat wash steps with 1ml of 1X PBS two more times.
- 3.1.3) Incubate an equimolar concentration of Pan-pRab, UDD1, UDD2 and 8G10 antibodies to give a concentration of 50 ng/µl of slurry. (Example, for 24 samples with two round of Immunoprecipitation, you require 24µg of Pan-pRab + 24µg of UDD1 + 24µg of UDD2 + 24µg of 8G10 in a total of 480 µl of protein A/G slurry which will give 50ng/µl concentration)
(Note: Due to the high binding capacity of Protein A/G beads all four antibodies can be simultaneously conjugated with 25 µg can be multiplexed to conjugate with 500 µl of Protein A/G beads slurry. For 24 samples 12 µg of each of the antibodies are required i.e. 0.5 µg per sample).
- 3.1.4) Add extra 1X PBS to a final volume of 1 ml.
- 3.1.5) Incubate at 4°C for 3 h on an end-to-end rotator.
- 3.1.6) Post incubation, centrifuge at 2,000 g for 2 minutes at 4°C and collect the supernatant to be used as a flow-through fraction to confirm by SDS-PAGE that the antibodies are bound to the beads completely.
- 3.1.7) Wash with 1 ml of 1X PBS as described above for a total of 3 times. Now add 1X PBS to a final volume of 480 µl to achieve an antibody concentration of 50 ng/µl of beads slurry. This mixture is stable at 4 °C for up to two months. For a longer storage add 0.03% (by vol) sodium azide.

3.2) Preparation and storage of heavy/light stable synthetic peptides:

- 3.2.1) Heavy/light synthetic peptides generally come in a lyophilized form at a net amount of 1 nmol. (Our lab has pRab1, pRab3, pRab8, pRab10, pRab35 and pRab43 and LRRK2 pS910 and pS935 and total LRRK2 heavy synthetic peptides)
- 3.2.2) Dissolve peptides in 100 µl of reconstitution buffer (3% Acetonitrile (by Vol) in 0.1% formic acid (by Vol) in HPLC-grade water to achieve 10 pmol/µl. Leave tubes on ice for about 20 min and then sonicate using a water bath sonicator for five min.

- 3.2.3) Vortex and centrifuge tubes using benchtop microfuge for 30 seconds.
- 3.2.4) Now prepare dilutions using reconstitution buffer as required e.g. 10pmol/ μ l, 1pmol/ μ l, 100 fmol/ μ l and 10fmol/ μ l. From these stocks prepare an equimolar concentration of all six pRabs+3 LRRK2 peptides to get 1pmol/ μ l or 100fmol/ μ l or 10fmol/ μ l concentrations to be Spiked into the peptide mixture just before immunoprecipitation. (For example, to achieve 1000 fmol/ μ l of an equimolar concentration of 200 μ l final volume, take 20 μ l from each peptide (6 Rabs+3LRRK2 peptides) from a stock of 10 pmol/ μ l to a total volume of 160 μ l and then add 40 μ l of reconstitution buffer to achieve a final concentration of 1000fmol/ μ l in 200 μ l volume. Store these stocks in -80 ° freezer).
- 3.2.5) Maintain 10 fmol/ μ l as working stocks so that 10 μ l should give 100 fmol net amount in immunoprecipitation reaction (Note: To avoid pipetting errors it is recommended to maintain the desired amount that will be spiked into the peptide mixture to be at a minimum of 10 μ l. e.g. For 100 fmol add 10 μ l of 10 fmol/ μ l stock. If you intend to Spike 50 fmol amount, then add 10 μ l of 5fmol/ μ l stock).

3.4) Protocol for Peptide immunoprecipitation:

- 3.4.1) Spin the lyophilized peptides from Step 2.3.24, at 17,000g at room temperature for 2 min to ensure everything settles down at the bottom of the tube. Add 1 ml of 1X IAP buffer, keep samples on water bath sonication for 10 min. The pH should be around 7.2. If the digest is still acidic, adjust the pH by adding 5 μ l of 1 M Tris pH 12.0. Centrifuge at high speed at room temperature for 10 min to remove any insoluble traces and transfer supernatant to new 1.5 ml Eppendorf tube. (Note: Use pH strips that have a fine gradient scale from pH 1.0 to pH 10, to adjust the pH)
- 3.4.2) Keep samples on ice. In the meantime, aliquot required amount of antibody slurry from the stock as prepared in Step 3.1.7. (Note: For each peptide immunoprecipitation containing up to 100-150 μ g tryptic peptide we use 10 μ l of slurry containing 0.5 μ g of Pan-pRab + 0.5 μ g UDD1 + 0.5 μ g UDD2 + 0.5 μ g 8G10 antibody. Also, add antibody cocktail using non-autoclaved cut tips. (Note: Due to low abundance of pRabs+LRRK2 pS910, pS935 in brain the recommended peptide amount to be used is >600 μ g peptide digest derived from 1 to 1.5mg starting lysate)
- 3.4.3) Add 100 fmol amount of an equimolar heavy stable peptides derived from step 3.2.5 (Note: We use an equimolar mixture of 10 μ l of 10fmol/ μ l stock to achieve 100 fmol amount containing six pRabs and LRRK2 pS910, pS9135 and unmodified peptide).
- 3.4.4) Incubate the mixture at 4°C for 2 h on an end-to-end rotator.

- 3.4.5) After the incubation, centrifuge at 1,500 g for 2 min at 4°C. Collect the flow through and transfer it to new protein lo-binding 1.5 ml Eppendorf tubes for second round of immunoprecipitation.
- 3.4.6) Keep beads on an ice-block on ice until the second round of immunoprecipitation is completed.
- 3.4.7) To the flowthrough from the 1st round of immunoprecipitation (Step 3.4.5), add another 10 µl of antibody slurry containing 0.5 µg of Pan-pRab + 0.5 µg UDD1 + 0.5 µg UDD2 + 0.5 µg 8G10 antibody and continue with the second round of immunoprecipitation for another 2 h at 4 °C on an end-to-end rotator.
- 3.4.8) Centrifuge at 1,500 g for 2 min at 4°C. Collect the flow through and transfer it to new 1.5 ml Eppendorf tubes. (Note: This flow through can also be used for subsequent enrichment of a different analyte or second round of IP or phosphopeptides using TiO₂)
- 3.4.9) Add 400 µl of 1X IAP buffer to the beads and carefully transfer it to the beads that are kept on ice from 1st round of immunoprecipitation in step 3.4.6.
- 3.4.10) Add another 600 µl of 1X IAP to the beads and gently wash tubes by inverting up-and-down for about 20 times (Note: This can be easily done by keeping samples on an ice-block and cover it with silver foil. Now complete wash by inverting up-and-down steps for all tubes at once. Use this technique for all subsequent wash steps.)
- 3.4.11) Centrifuge tubes at 1500g at 4°C for 3 min. Place tubes on ice-block and allow the beads to settle for at least two min before aspirating the supernatant.
- 3.4.12) Repeat washing by adding 1 ml of ice-cold 1 X IAP buffer and centrifuge as in Step 3.4.11 and follow the same conditions for all subsequent wash steps.
- 3.4.13) Wash the beads with 1 ml of ice-cold 1X PBS, Centrifuge and repeat this wash for once more.
- 3.4.14) Wash beads with 1 ml of ice-cold Milli-Q water, centrifuge.
- 3.4.15) After the final wash, centrifuge and aspirate all water and add 50 µl of 0.2% (by vol) TFA to the beads. Gently tap for proper mixing and incubate at room temperature for about 10 min on a Thermo mixer at 1250 rpm.
- 3.4.16) Centrifuge at 1,500 g for 2 min at room temperature and collect the eluate into a new collection tube with a gel loader tip
- 3.4.17) Repeat the elution step by adding another 50 µl of 0.2% (by vol) TFA followed by incubation at room temperature for 10 min and centrifuge and collect the supernatant into the previous collection tube and proceed with C18 stage-tip clean up (Section 2.5).

3.5) C18 Stage-tip protocol:

- 3.5.1) Prepare single layer of C18 stage-tip using 16-gauge syringe needle as shown in figure **Figure2**
- 3.5.2) Add 80 µl of 100% (by vol) Acetonitrile to the C18 stage-tip (**Note this is required to activate the C18 resin**).
- 3.5.3) Centrifuge at 2000g for 2 min at room temperature. Discard flow through
- 3.5.4) Add 80 µl Solvent-A1 0.1% (by vol) TFA (by vol) (**Note this is required to equilibrate the C18 resin**).
- 3.5.5) Centrifuge at 2000g for 2 min at room temperature. Discard flow through and repeat this step.
- 3.5.6) Load the acidified peptide digest from step **3.3.17** to Stage-tip.
- 3.5.7) Centrifuge at 1500g for 5 min at room temperature.
- 3.5.8) Reapply the flow through to the C18 stage-tip column and repeat this step.
- 3.5.9) Add 80 µl of solvent-A1 (0.1% (by vol) TFA v/v), and centrifuge at 2,000g for 2 min at room temperature. Discard flow through. Repeat again.
- 3.5.10) Elute peptides absorbed to C18 column by placing the stage-tips into **new** 1.5 ml low binding tubes.
- 3.5.11) Add 30 µl of Elution buffer (Solvent B1- 50% (by vol) acetonitrile in 0.1% (by vol) TFA).
- 3.5.12) Centrifuge at 1,500g for 2 min. Add another 30 µl of Elution buffer (Solvent B1- 50% (by vol) acetonitrile in 0.1% (by vol) TFA) and repeat the centrifugation.
- 3.5.13) Immediately snap freeze on dry ice the eluates and vacuum dry the samples using a Speed-Vac
- 3.5.14) Store samples in -80 freezer until the LC-MS/MS analysis

